

Hanson He

Waterloo, ON | s296he@uwaterloo.ca | +1 437-385-5777 | [linkedin.com/in/shengyuan-he](https://www.linkedin.com/in/shengyuan-he) | github.com/HansonHe-UW

Education

University of Waterloo, Waterloo, ON

Sep 2026 – Apr 2031

Bachelor of Applied Science (BASc), Honours Electrical Engineering, Co-op Program

- **President's Scholarship** (Jun 2025) — Awarded for academic excellence with an admission average between 90% and 94.9%.
- **Relevant Coursework:** Linear Algebra, Calculus 1 for Engineering, Programming Fundamentals in C++, Classical Mechanics.

Technical Skills

- **Embedded & Hardware:** Arduino, Lab Instrumentation (Multimeter, Oscilloscope), Embedded C
- **Programming Languages:** Python, C/C++ (Embedded), JavaScript/TypeScript, SQL
- **AI / Data & CV:** Gemini API, sentence-transformers, ChromaDB, OpenCV, YOLOv8
- **Web & Backend:** FastAPI, REST APIs, WebSockets, Node.js
- **Frontend:** React, Vite, Tailwind CSS, HTML/CSS
- **Tools & Systems:** Git, Linux/Unix, Bash/Shell, Firebase

Projects

SpatialVCS | FastAPI, React 19, WebSockets, YOLOv8, Gemini 2.5 Flash

February 2026

Full-Stack Developer (Team of 3) — **Best Beginner Hack @ CTRL+HACK+DEL 2.0**

- Built “Git for Reality” in 36h (**FastAPI + React 19**, team of 3) — transforming physical spaces into searchable snapshots for **visually-impaired users**.
- Designed a “Physical Diff” algorithm with centroid-based tracking, using mobile phones as **edge sensors** streaming camera frames over **WebSocket** for **YOLOv8** detection.
- Enabled privacy-preserving natural language queries via **ChromaDB RAG + Gemini 2.5 Flash**, storing no raw video frames.

Wildfire Tactics | TypeScript/React, Deck.gl, Mapbox, LLM Agents

Jan 2026 – Feb 2026

Software Engineer — Hack Canada 2026 (Team of 4)

- Built a wildfire evacuation simulator in **TypeScript/React** (team of 4) modeling fire spread, road-constrained routing, and **multi-agent** resident/guide behavior on **Deck.gl + Mapbox**.
- Implemented **shared flow-field routing** with threat-aware path costs (fire intensity, roadblocks) for efficient crowd evacuation across hundreds of agents.
- Integrated **LLM-based guide agents** with function calling to select safe points from live simulation state, with secure key handling via a **Vite proxy**.

Experience

IT Support & Automation Intern

Apr 2025 – Jul 2025

Hunan Shengpeng Electric Power Technology Co., Ltd. — Changsha, China

- Automated new-device setup (app installs, printer/VPN config) via batch/shell scripts, eliminating repetitive manual steps.
- Tracked incidents and assets in a shared Sheet to streamline handoff and reduce repeat tickets.
- Authored standardized checklists for common fixes (Wi-Fi, shared drives, permissions), improving first-try resolution.
- Troubleshoot Windows/macOS endpoints and network connectivity across a multi-device small office.

Extracurricular Activities

UW Computer Science Club — Event Coordinator

May 2026 – Present

- Organize termly events across technical (workshops, Code Parties, industry talks) and community (trivia nights, picnics) streams for UW CS undergraduates.
- Coordinate logistics, speaker/sponsor outreach, and cross-channel promotion (Discord, LinkedIn) for events drawing 50+ attendees.